

Task for the preparation of a Master Thesis

Course: Master NES
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Title: **FPGA-based Real-Time Learning Architecture with Partial Reconfiguration for Spiking Neural Networks**

Objectives of work

Spiking Neural Networks (SNNs) are brain-inspired models that process information using discrete spike events instead of continuous values. This makes them suitable for low-power, real-time implementations on hardware such as FPGAs. However, efficient online learning in SNNs is still difficult to achieve in hardware.

The main goal of this thesis is to investigate efficient and flexible real-time learning in SNNs on FPGA platforms. It focuses on existing digital SNN designs with online learning, with emphasis on the SDSP (Spike-Driven Synaptic Plasticity) algorithm. The SDSP rule offers a hardware-friendly approach to learning through local update mechanisms. Additionally, the thesis will explore Partial Reconfiguration (PR) techniques to enable dynamic changes of the training algorithm, while the rest of the system continues running on the FPGA.

Focus of work

- Conduct a literature review on:
 - Digital implementations (FPGA or ASIC) of SDSP training algorithm.
 - FPGA-based SNN implementations with PR-based real-time training algorithm.
- Design and optimize an architectural model for the SDSP algorithm.
- Implement the model into RTL Code, simulate and validate it.
- Integrate the training module into a simplified SNN design (taking advantage of an open-source SNN framework).
- Make the training dynamically reconfigurable (Apply PR to the complete training block).
- Test it on FPGA and get (both with and without PR) the following metrics:
 - Resource utilization, power consumption, latency, maximum frequency, accuracy, and for PR, reconfiguration time.
 - (Optional): Convergence speed for learning.
- Comparison with other digital SNN architectures:
 - With real-time SDSP training possibility.
 - with PR-based real-time training possibility.
- (Optional) Extend the simplified SNN to support more neuron models and/or architecture settings.

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